

How to enhance online reputation? An empirical study from a joint customer–firm perspective

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Abstract

Online reputation is critical for hotel firms, as evidenced by consumer ratings in online media. The present study aims to develop insights into the antecedents of the hotel firm's online reputation and thus make a dual contribution to the literature by (a) examining the strategies implemented via online media as antecedents and (b) combining the firm's perception of the value it has delivered with the market's perspective. A quantitative empirical study on hotel firms and their consumers was conducted. The results show that the firm's perception of the value it has delivered ("firm perceived value")—the antecedents of which are strategic online reputation-management and the strategic use of revenue-management—has a positive and significant effect on online reputation (as measured from the customer perspective), and that strategic online reputation-management has a positive and significant effect on the strategic use of revenue-management.

Keywords

Online reputation, revenue-management, firm perceived value, strategic online reputation-management, hotel firms

Introduction

The proliferation of social media, which enable tourists to share their experiences with each other, has rendered the role played by opinions posted online about tourism suppliers increasingly important for the tourism sector (Su et al., 2022). These types of opinions are usually summarized in a numerical value or rating, which constitutes the firm's online reputation (e.g. Anderson and Lawrence, 2014; Rodríguez-Díaz et al., 2019; Yang et al., 2018). Online reputation plays an essential role in consumer decision making in the tourism sector (Tsao et al., 2019) because it helps consumers to select service providers (Bigné et al., 2020) by taking the effort out of reaching decisions—hence, consumers often place their trust in it (Yang et al., 2018).

The literature has highlighted the effect of online reputation both on consumer behavior

(e.g. Gavilán et al., 2018; Wen et al., 2021) and on the performance of tourism firms (e.g. Nieto-García et al., 2019; Viglia et al., 2016) as a source of competitive advantage (Cezar and Ögüt, 2016), among hotel firms in particular. However, despite the importance of online reputation, there is little scholarship analyzing what firms might do, strategy-wise, to enhance it, resulting in calls for more research on this question (De Pelsmacker et al., 2018; Serra-Cantallos et al., 2020). Given that a firm's online reputation reflects the market's assessment of the service it

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delivers (Hu and Yang, 2021; Liang et al., 2017), it is of interest to examine whether the firm's own perception of the value it delivers to customers—that is, “firm perceived value” (hereafter, FPV)—influences its online reputation. We contend that this is a critical factor, as is only by taking into account both perspectives (internal/firm and external/customer) that the firm will be able to deliver the value that is expected and sought by the market (O'Cass and Ngo, 2011). The present study contributes to a more nuanced understanding of this dual perspective.

To strategically develop an offer of superior value to the market, firms are increasingly turning to online media as a source of market intelligence (Duncan et al., 2019). Online reputation-management and revenue-management are two such strategies implemented by hotel firms via online media that draw on the market intelligence these channels provide (Aureli and Supino, 2017; Xu et al., 2019b). On the one hand, strategic online reputation-management involves obtaining and analyzing information from the market via online media (De Pelsmacker et al., 2018), which firms can then use to improve the service they deliver to the market (Pérez-Aranda et al., 2019). On the other hand, revenue-management can use this information to design value offerings that can be uniquely customized to each segment (Xu et al., 2019b). Thus, strategic online reputation-management encourages hotel firms to make more strategic use of revenue-management (Alrawadieh et al., 2021; Xu et al., 2019b). However, the literature to date is yet to empirically analyze the effect of strategic online reputation-management and the strategic use of revenue-management on FPV, and the effect of strategic online reputation-management in terms of facilitating a more strategic use of revenue-management. That said, authors including Bore et al. (2017) and Niu and Fan (2018) highlight the need to study the effects of strategic online reputation-management, while Denizci Guillet and Mohammed (2015) and Selmi and Chaney (2018) make the same recommendation with respect to revenue-management.

The overarching aim of the present study is therefore to identify how the online reputation of firms can be improved. We propose and validate a model to determine whether: (a) FPV influences the firm's online reputation, as judged from the external market's point of view (according to online consumer ratings); (b) strategic online reputation-management and the strategic use of revenue-management influence FPV; and (c)

strategic online reputation-management influences the firm's strategic use of revenue-management.

Literature review

Online reputation as a source of competitive advantage for hotel firms

Online reputation can be broadly defined as the combined sentiment conveyed by the content that consumers freely post about a firm or brand via online media (Cillo et al., 2021), and the ever-growing prevalence of social media has heightened its importance (Su et al., 2022). Social media can be understood as a set of tools, applications, platforms, and online media that enable people to create and exchange content in peer-to-peer communication channels, ranging from social networks and consumer review sites to online content communities, wikis, and other types of information and communications technology (ICT) resources (Mirzaalian and Halpenny, 2019; Kaplan and Haenlein, 2010; Zeng and Gerritsen, 2014).

Among the various tourism-related social media, consumer review sites are particularly important. These sites are platforms on which consumers can post content about tourism services, and they can be community-based (such as TripAdvisor or Yelp) or transaction-based (such as the online travel agencies Expedia.com or Booking.com) (Mirzaalian and Halpenny, 2019). On this kind of site, generally, the evaluation made by consumers after consuming a service is summarized in a numerical value or rating, which can be considered equivalent to the firm's online reputation (e.g. Anderson and Lawrence, 2014; Rodríguez-Díaz et al., 2019; Yang et al., 2018).

Online reputation is particularly important in the tourism context, given that the intangible, perishable, and experiential nature of tourism products renders it difficult for consumers to evaluate them prior to purchase (Yang et al., 2018). Thus, to mitigate their uncertainty, consumers seek impartial and reliable information to build a true image of what the product is like, primarily sourcing that information from the aforementioned online sites that make it easy for consumers to share their experiences with each other (Bortoluzzi et al., 2020). These sites have proliferated, leading online reputation to play an increasingly crucial role in the tourism sector (Nieto-García et al., 2019).

Table 1. Studies dealing with the effects of online reputation in the hotel industry context.

Studies that theoretically support the influence of online reputation on consumer behavior	
Argument	Study
Online reputation is the first impression to which users are exposed when they search for information about a hotel	O'Connor (2010)
It reduces the time consumers need to spend searching for information	Nieto-García et al. (2019)
It helps consumers to establish a selection criterion (the hotels with the best online reputation are the most popular)	Gavilán et al. (2018); Casalo et al. (2015)
It is a signaling mechanism that influences consumers' purchase decisions	Liang et al. (2017)
It replaces other forms of word-of-mouth communication regarding hotel quality	Gavilán et al. (2018)
It acts as a recommendation mechanism for future consumers, thus affecting the offline reputation of the hotel firm	Radojevic et al. (2017)
Consumers are willing to pay premium prices for hotels with a good online reputation	Mellinas et al. (2016)
It is crucial for the hotel firm in terms of visibility	Nieto-García et al. (2019)
It matches the opinions of consumers captured through offline media	Smironva et al. (2020)
It can help build brand loyalty	Gavilán et al. (2018)
Studies that demonstrate empirically the influence of online reputation on consumer behavior	
Variable influenced by online reputation	Studies
Booking intention	Casalo et al. (2015); Hu and Yang (2020); Park et al. (2017); Sparks and Browning (2011); Wen et al. (2021)
Choice of hotel	Noone and McGuire (2013)
Brand attitude	Casalo et al. (2015)
A change in brand attitude	Lee and Ro (2016)
Perceived value	Noone and McGuire (2013, 2014)
Perceived quality	Wen et al. (2021)
Hotel booking consideration	Gavilán et al. (2018); Hu and Yang (2020)
Trust in the hotel	Sparks and Browning (2011)
Willingness to pay	Nieto-García et al. (2017)
Studies that demonstrate empirically the influence of online reputation on hotel firms' business performance indicators	
Variable influenced by online reputation	Studies
Bookings	Ögüt and Onur Tas (2012); Masiero and Nicolau (2016)
Revenue per available room	Kim et al. (2015); Nieto-García et al. (2019)
Occupation rate	Kim et al. (2015); Viglia et al. (2016)
Market share	Duverger (2013)
Financial performance	Anderson and Lawrence (2014)
Income	Zervas et al. (2017)
Income from online transactions	Torres et al. (2015)
Conversion rate	Cezar and Ögüt (2016)

Source: The authors.

The literature emphasizes the importance of the firm's online reputation (understood as the rating given to it by previous consumers) in terms of consumer behavior and business performance in the tourism sector (Bigné et al., 2020), in particular, the hotel industry. Table 1

summarizes the studies that: (a) theoretically support the influence of online reputation in helping consumers in their decision-making process and helping hotel firms achieve visibility in the market (e.g. Gavilán et al., 2018; Nieto-García et al., 2019; Smironva et al.,

2020); (b) empirically demonstrate the influence of online reputation on consumer-behavior variables, such as hotel-booking intention or brand attitude (e.g. Casaló et al., 2015; Hu and Yang, 2020; Wen et al., 2021); and (c) empirically demonstrate the influence of online reputation on hotel firm performance (e.g. Kim et al., 2015; Masiero and Nicolau, 2016; Viglia et al., 2016). Given its importance, then, various studies highlight the need to determine the antecedent variables that can influence online reputation (De Pelsmacker et al., 2018; Xu et al., 2019a).

Firm perceived value and its effect on online reputation

From the business perspective, each firm has its own perceptions of the value it is delivering to the market (Martelo-Landroguez et al., 2015), which we term FPV. This value constitutes a decisive step toward ensuring that the consumer ultimately attaches a high perceived value to the services it has consumed (O'Cass and Ngo, 2011).

The role of managers in delivering superior value to the market has been widely recognized both conceptually and theoretically in the literature. In fact, in the tourism field, due to the physical separation between the place of decision making and that of consumption, the study of the perceived value delivered by the firm is even more necessary (Gallarza et al., 2015; Nasution and Mavondo, 2008). Furthermore, when a firm implements an internal strategy to deliver greater value to the market, this may or may not be perceived or valued by consumers (Polo-Peña et al., 2016). However, if firms take into account both perspectives (their own and that of the market), this may reduce the gap between what managers believe they deliver and what consumers actually experience (Radder et al., 2016).

Ultimately, it is the relationship between FPV and the value perceived by the market that can generate a positive impact on consumer behavior (Martelo-Landroguez et al., 2015; Nasution and Mavondo, 2008). Hence, there are studies that provide empirical evidence showing that superior FPV enables the firm to achieve (a) superior performance (Chen and Myagmarsuren, 2013) and (b) a positive effect on variables of consumer behavior such as satisfaction (Gallarza et al., 2015) and, of course, consumer perceived value (Martelo-Landroguez et al., 2015; O'Cass and Sok, 2013, 2015).

In light of the above findings from the literature, and given that the online-reputation variable reflects how the market values the service offered by the firm (Hu and Yang, 2021; Liang et al., 2017), it is just as imperative to take FPV into account (Pérez-Aranda et al., 2018). Taking this dual perspective, it is also vital to study how FPV may influence the hotel firm's online reputation (as assessed externally by the market, in the form of the rating that consumers give the hotel via online media). To the best of our knowledge, the literature to date has not examined this relationship, hence, this is one of the primary contributions of the present study.

The following research hypothesis is therefore proposed:

- H1.** FPV has a positive and significant effect on the online reputation of the firm.

Strategic online reputation-management and strategic use of revenue-management to improve firm perceived value

Online media provide an ideal platform from which to develop marketing strategies geared to obtaining, managing, and leveraging market intelligence to provide greater customer value (Duncan et al., 2019). On this point, the literature highlights the major potential of strategic online reputation-management (e.g. Lunkes et al., 2020) and revenue-management (e.g. Selmi and Chaney, 2018).

Online reputation-management consists of monitoring, evaluating, and addressing the sentiments expressed in user-generated content about the firm in online media to uphold its online reputation (Cillo et al., 2021). Various authors contend that strategic online reputation-management in the hotel context needs to go a step further, arguing that it should be considered a process (Pérez-Aranda et al., 2019) designed to identify the firm's strengths and areas for improvement (Levy et al., 2013). To achieve this, hotel firms should harness this process to actively encourage consumers to generate opinions about the firm in online media (De Pelsmacker et al., 2018; Levy et al., 2013), should give prompt, personalized responses to those opinions (Liao et al., 2022), and should analyze them systematically (Niu and Fan, 2018) in the pursuit of continual improvement (Baka, 2016).

Thus, one of the essential aspects of strategic reputation-management is to encourage the

generation of online content about the firm by consumers and to analyze that content (De Pelsmacker et al., 2018; Levy et al., 2013). Systematically gaining access to online reviews is the best way for firms to gauge the satisfaction or dissatisfaction of consumers, their demands, expectations, and needs, and this approach also helps identify opportunities for service improvement (Lunkes et al., 2020; Pérez-Aranda et al., 2019). This, in turn, enables the firm to gain consumer trust (Nasution et al., 2011) and achieve differentiation from the competition.

In short, having mechanisms in place to systematically obtain a deep understanding of the preferences expressed by consumers enables firms to offer superior value to them (Garrido-Moreno et al., 2018; Nasution and Mavondo, 2008). It follows, then, that the generation of market intelligence by means of strategic online reputation-management will enable the firm to deliver more value to the market (Nasution et al., 2011). In turn, it is logical to expect that the hotel firm's strategic management of its online reputation may affect FPV (Pérez-Aranda et al., 2019).

However, research dealing with online reputation-management has tended to focus solely on tactical aspects, with the exception of the study by Pérez-Aranda et al. (2019), which adopts a strategic vision of online reputation-management and measures its influence on business profits. Scholars have therefore noted the need to analyze the effects of strategic online reputation-management (Bore et al., 2017; Niu and Fan, 2018), and its effect on FPV is of particular interest. The following research hypothesis is therefore proposed:

- H2.** Strategic online reputation-management has a positive and significant effect on FPV.

Meanwhile, revenue-management, which can be defined as the process of assigning the right kind of capacity to the right type of customer, at the right price, at the right time, through the right distribution channel to maximize the firm's revenue or performance (Denizci Guillet and Mohammed, 2015) was traditionally regarded as a more tactical and "inventory-centric activity" (Noone et al., 2011). But, thanks to the development of online media, it has now adopted a more strategic perspective (Erdem and Jiang, 2016) characterized by a focus on the consumer and the generation of demand. This approach identifies the right offers and prices that effectively

deliver value to consumers (Rodríguez-Díaz et al., 2019).

Revenue-management uses market intelligence to understand market behavior (Guillet, 2020). The use of online media helps in the acquisition of this market intelligence, as the views that consumers express via such media can be integrated into the hotel firm's revenue-management system, providing insights into their preferences, perceptions, and behaviors (Xu et al., 2019b). In this way, online media can help firms with the decision making that strategic revenue-management requires, by providing them with a better understanding of consumer behavior (Wang et al., 2015).

Given that revenue-management will not be effective if it fails to deliver value to the market (Noone and McGuire, 2013), its strategic focus calls for hotel firms to employ online media to understand the value expected by the market at a given price point (Noone et al., 2011) and thereby ensure that the value actually delivered to the market is as aligned as possible with the value expected by consumers (Rodríguez-Díaz et al., 2019). What is more, the incorporation of online media in revenue-management will facilitate the delivery of personalized value offers (Noone et al., 2017) thanks to the creation and maintenance of relationships with consumers that these media facilitate (Wang et al., 2015), as well as providing a better understanding of hotel-booking behavior (Xu et al., 2019b).

Consequently, the strategic use of revenue-management will enable the firm to provide an offer of superior value to the market thanks to the sound management of room and service availability and the setting of prices that are effectively matched to different consumer segments (Noone et al., 2011; Xu et al., 2019b).

The present study, then, aims to address the potential future research proposed by Wang et al. (2015), who evidenced the need to conduct studies on the revenue-management opportunities presented by online media. More specifically, the study seeks to advance toward a better understanding of the effect of the *strategic* use of revenue-management on FPV, responding to the call for future research proposed by Denizci Guillet and Mohammed (2015). The following research hypothesis is therefore proposed:

- H3.** The strategic use of revenue-management has a positive and significant effect on FPV.

Both strategic online reputation-management and revenue-management benefit from the possibilities offered by online media to obtain information from consumers and insights into demand that can be used by the firm to generate market intelligence (Duncan et al., 2019). Strategic online reputation-management generates market intelligence that enables hotel firms to ascertain the needs and preferences of consumers (Aureli and Supino, 2017), while revenue-management uses that market intelligence to adjust its offers to market expectations and continuously offer superior value (Selmi and Chaney, 2018). Those responsible for revenue-management in the firm rely on strategic online reputation-management to understand consumer perceptions, expectations, and behaviors, and, based on this understanding, determine the right offers for the right consumers (Alrawadie et al., 2021), thereby improving the firm's revenue-management system (Xu et al., 2019b).

Xu et al. (2019b) make inroads into studying the effect of strategic online reputation-management on the strategic use of revenue-management. These authors demonstrate the positive effect of drawing on user-generated online-media content in terms of enhancing the sophistication of the revenue-management

system. This leads us to suggest that, similarly, strategic online reputation-management can exert a positive effect on the strategic use of revenue-management. In view of these findings, the following research hypothesis is proposed:

- H4.** Strategic online reputation-management has a positive and significant effect on the strategic use of revenue-management.

The proposed research model is shown in Figure 1.

Methodology

Research population and sample

We conducted an empirical study with a dual focus. On the one hand, we examined the perspective of firms operating in an industry in which a strong online reputation is particularly important and strategies based on market intelligence generated via online media—specifically, online reputation-management and revenue-management—play a major role. We selected the Spanish hotel industry for this purpose. On the other hand, we also examined the perspective of consumers of hotel services.

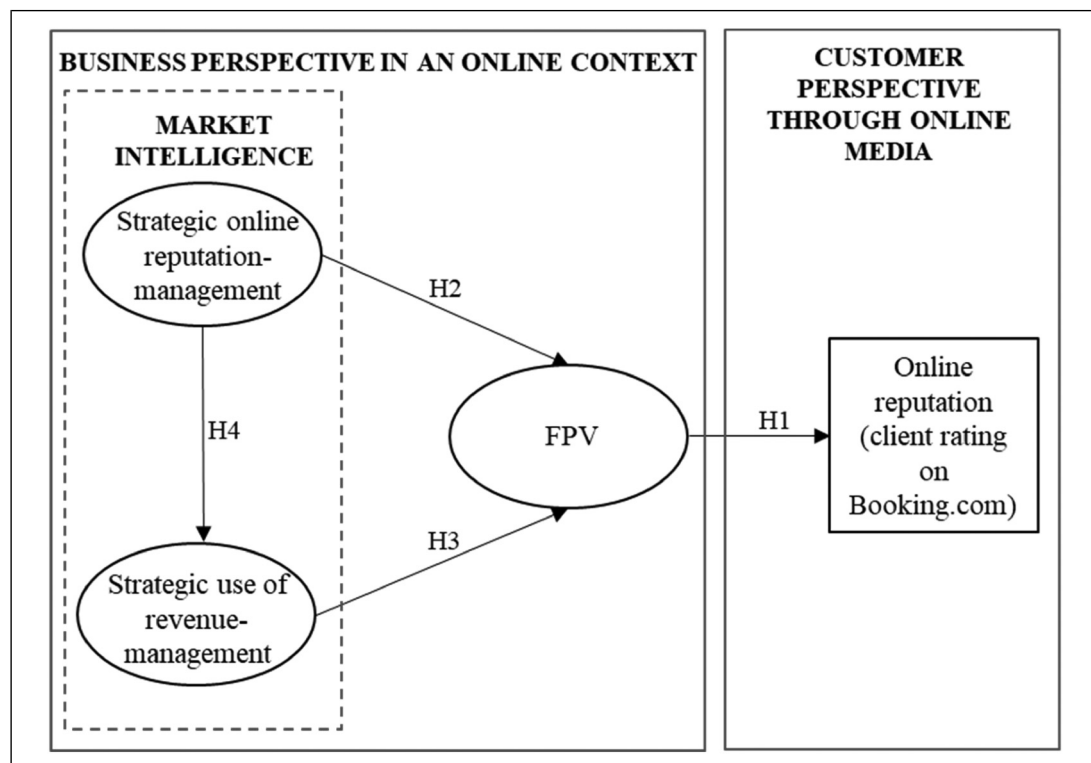


Figure 1. Proposed research model.

Our rationale for selecting the Spanish hotel industry as the context for this study was that (1) being a highly competitive service industry, online reputation plays a very significant role (Smironva et al., 2020); (2) hotels have placed increasing emphasis on strategic online reputation-management (Baka, 2016) and revenue-management (Denizci Guillet and Shi, 2019) in recent years; and (3) this industry is a critical economic driver in Spain (Garrido-Moreno et al., 2018), which is the second most-visited destination in the world (WTO, 2019). We focus, in particular, on chain hotels with a presence in Spain, because chains tend to demonstrate more innovation and professionalism than independent establishments and have access to better resources with which to develop marketing plans, gather data, and integrate their ICTs (Flecha Barrio et al., 2016).

To construct the sampling frame, we drew on the ranking published by Hosteltur (www.hosteltur.es), which is considered the primary ranking for identifying Spanish hotel chains (García-Muina et al., 2020). As this resource exclusively captures Spanish-owned chains, we completed the sampling frame by conducting a search on Booking.com to identify other hotel chains with a presence in Spain but foreign-owned, as well as any other chains not included in the Hosteltur ranking. Booking.com was chosen for this purpose because it is the world's leading website for hotel reservations (Rita et al., 2022). We identified a total of 330 hotel chains with a presence in Spain and we opted to access the entire universe—a tactic suggested by various authors to deal with small sampling frames (Hair et al., 2018; Flecha Barrio et al., 2016).

For the purpose of data collection, which took 4 months, the online marketing managers of hotels belonging to hotel chains were contacted by telephone; and, subsequently, a questionnaire

was administered to them via email. Some 137 hotels belonging to chains responded, representing 33% of the universe of chains with a presence in Spain. This can be considered a relatively high sample considering that it comprises hotel firms (e.g. Nieves and Quintana, 2018).

Turning to the market perspective, this was captured through the consumer opinions posted publicly on the Booking.com profile of the hotels that responded to the survey. Booking.com is the leading website for online hotel ratings, hosting 39% of online opinions worldwide and an even higher percentage in the case of Europe only (Martín-Fuentes and Mellinas, 2018). Figure 2 shows the methodology used to obtain the data; and Table 2 sets out the sample characteristics.

Measurement scales

Respondents were asked to assess various items on the questionnaire relating to the constructs of strategic online reputation-management, strategic use of revenue-management, and FPV. These items were adapted from various studies in the tourism field and were measured on 7-point Likert scales (Supplemental Appendix).

To measure strategic online reputation-management, we used a scale based on extant studies: that of De Pelsmacker et al. (2018), in the hotel context, and that of Marchioria et al. (2013), in the tourism context. This scale adopted here measured the degree to which the hotel firm encourages consumers to generate opinions about it in online media, actively monitors and analyzes those opinions, and posts personalized responses to them.

To measure the hotel's strategic use of revenue-management, we selected a scale based on the work of Selmi and Chaney (2018), whose study was conducted in the hotel context, and adapted it to the online context.

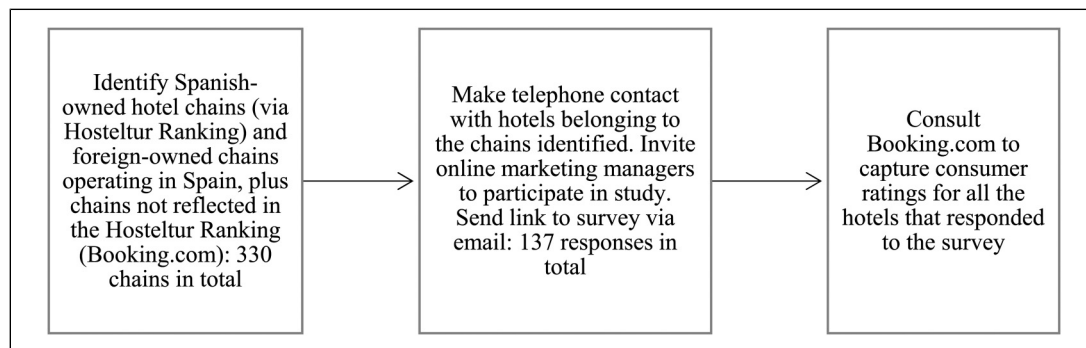


Figure 2. Data-collection methodology.

Table 2. Sample characteristics.

Aspect	Number of responses	Percentage of sample (n = 137)
Number of stars		
1-star	5	3.65
2-star	5	3.65
3-star	29	21.17
4-star	82	59.85
5-star	16	11.68
Number of years in operation		
Up to 5 years	27	19.71
6–10 years	17	12.41
Over 10 years	93	67.88
Gender of respondent		
Male	69	50.40
Female	68	49.60
Age of respondent		
18–29 years	17	12.41
30–44 years	87	63.50
45–65 years	33	24.09
Educational profile of respondent		
No university education	20	14.60
University education	117	85.40
Respondent's number of years in current post		
Up to 5 years	29	21.17
6–10 years	41	29.93
More than 10 years	67	48.90

This scale measures the degree to which the entire firm is oriented toward using online media for strategic revenue-management and to maximize long-term profitability by means of segmentation and dynamic management of the firm's capacity (Selmi and Chaney, 2018).

To measure FPV, we used a scale based on Nasution and Mavondo (2008), which has previously been applied in the tourism sphere (e.g. Frías-Jamilena et al., 2017). This scale measures the firm's perception of the value it offers consumers—that is, its perception of whether the benefits received by clients as a result of the service it delivers make up for the sacrifices they make in acquiring that service (Nasution and Mavondo, 2008).

Finally, to measure online reputation, we used the hotel's overall rating on Booking.com. Several studies empirically demonstrate that the Booking.com rating scale is valid and reliable to measure hotel online reputation (e.g. Rodríguez-Díaz and Espino-Rodríguez, 2018a, 2018b; Rodríguez-Díaz et al., 2019; Yacouel and Fleischer, 2012).

Results

In terms of procedure, first, we confirmed the adequacy of the measurement scales for the different variables, using structural equation modeling (SEM) with AMOS V.24 software. Next, using the same analysis technique, we tested the proposed relationships between the variables. SEM is appropriate for validating measurement scales and verifying causal relationships between constructs because it distinguishes between measurement instruments and the structural model, and it takes into account measurement errors in the estimation of the model (Hair et al., 2018).

Model fit and assessment of the measurement model

Given that the multivariate normality test of the variables was found to be significant, we used the maximum likelihood model combined with bootstrapping (Yuan and Hayashi, 2003). Regarding goodness-of-fit, the normed Chi-square produced a value of 1.15 and was therefore within the recommended range. As for overall model fit, the goodness of fit index (GFI) and the root mean square error of Approximation (RMSEA) values were both acceptable (0.94 and 0.03, respectively). The incremental fit measures of comparative fit index (CFI) (0.99), incremental fit index (IFI) (0.99), and Tucker-Lewis index (TLI) (0.99) were also adequate. The model fit can therefore be considered acceptable, in line with the recommendations of Hair et al. (2018).

Next, we evaluated the measurement model to check that the measurement scales correctly measured each of the latent variables (strategic online reputation-management, strategic use of revenue-management, and FPV). We examined the statistical significance of each load obtained between the latent variable and its indicator (which should be significant and have a standardized value >0.50, and ideally 0.70 or higher) (Hair et al., 2018). Table 3 shows that all the load values were above this threshold, the confidence interval did not include the value "0", and the values presented a significant p-value. We then estimated the individual reliability or R^2 of each item. All were found to be greater than, or close to, a value of 0.50, as recommended by the literature (Hair et al., 2018). Regarding the internal consistency check for each of the scales, both the composite reliability and the variance extracted presented values above, or very close

Table 3. Indicators for convergent validity and internal consistency of the scales.

Factor	Standardized loads, confidence interval, and p-value	Individual reliability (R^2), confidence interval, and p-value
Strategic online reputation-management (SORM) CR = 0.75; AVE = 0.43		
SORM1	0.73 (0.57; 0.86)**	0.53 (0.32; 0.73)**
SORM2	0.72 (0.54; 0.88)**	0.51 (0.29; 0.77)**
SORM3	0.60 (0.40; 0.78)**	0.36 (0.16; 0.60)**
SORM4	0.58 (0.37; 0.74)**	0.33 (0.14; 0.55)**
Strategic use of revenue-management (RM) CR = 0.92; AVE = 0.75		
RM1	0.71 (0.60; 0.81)**	0.50 (0.36; 0.65)**
RM2	0.91 (0.85; 0.96)**	0.83 (0.71; 0.92)**
RM3	0.92 (0.87; 0.96)**	0.85 (0.76; 0.92)**
RM4	0.91 (0.85; 0.95)**	0.83 (0.73; 0.91)**
Firm perceived value (FPV) CR = 0.90; AVE = 0.75		
FPV1	0.69 (0.50; 0.81)**	0.47 (0.25; 0.66)**
FPV2	0.94 (0.89; 0.97)**	0.88 (0.79; 0.94)**
FPV3	0.95 (0.88; 0.98)**	0.90 (0.77; 0.97)**

**p-value < 0.01.

AVE = variance extracted; CR = composite reliability.

to, the reference threshold (0.70 in the case of composite reliability and 0.50 in the case of variance extracted).

Finally, we evaluated the discriminant validity among the different variables included in the research model (Table 4). The correlations between variables were no greater than 0.80 (Bagozzi, 1994) and the confidence interval of the estimated coefficient did not include the value “1” (Anderson and Gerbing, 1988). Discriminant validity can therefore be verified. The results overall indicate that the scales used in the study were adequate for measuring each of the variables in the research model.

Testing the hypotheses

Next, the relationships between strategic online reputation-management, the strategic use of revenue-management, FPV, and online reputation were analyzed (Figure 3). In light of the results, the following points regarding the research hypotheses are of particular note:

H1 proposes that FPV has a positive and significant effect on the firm's online reputation. The results showed a statistically significant relationship between the two variables ($p < 0.05$). The direct effect was 0.19, with a confidence interval between 0.06 and 0.34. Thus, there is empirical support for this hypothesis, and we can conclude that FPV has a positive and significant effect on the firm's online reputation, as

expressed in the hotel's overall consumer rating on Booking.com.

H2 hypothesizes that strategic online reputation-management has a positive influence on FPV. The results showed a statistically significant relationship between the two variables ($p < 0.05$) and a direct effect of 0.35, while the confidence interval was between 0.09 and 0.58. Thus, this hypothesis finds empirical support, and we can conclude that strategic online reputation-management exerts a positive impact on FPV.

H3 proposes that the strategic use of revenue-management has a positive influence on FPV. The results showed a statistically significant relationship ($p < 0.01$), with a direct effect of 0.39 and a confidence interval between 0.19 and 0.60. Therefore, this hypothesis also finds empirical support, and we can conclude that the strategic use of revenue-management has a positive impact on FPV.

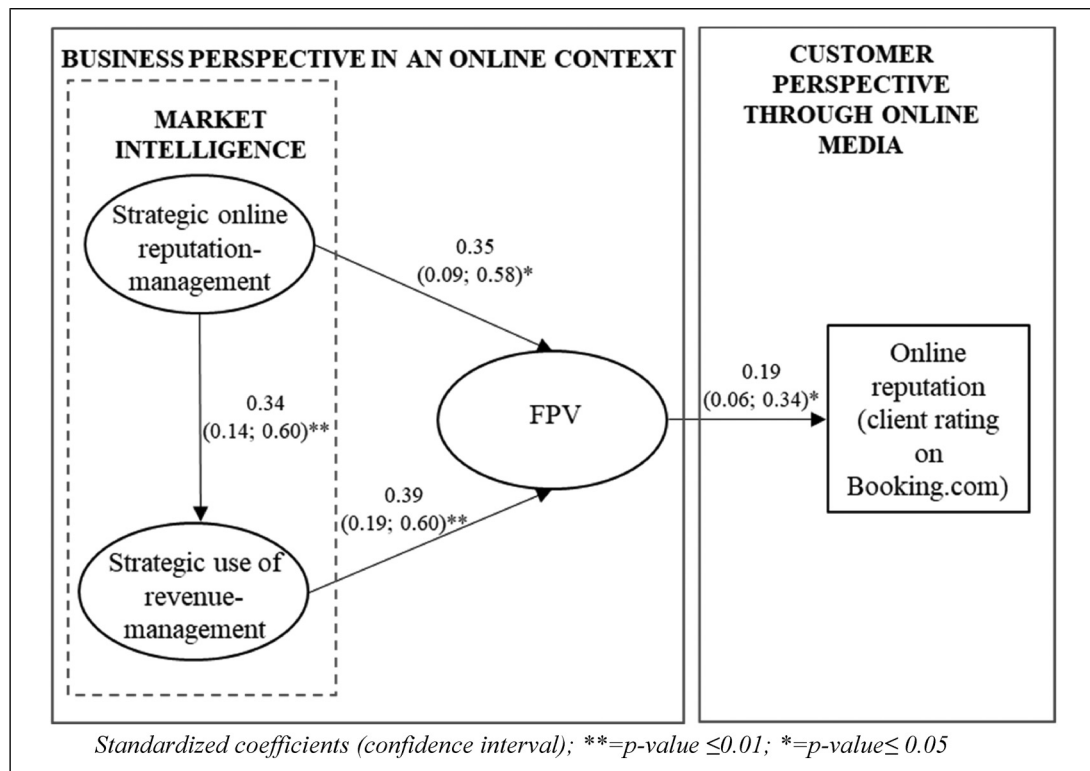
H4 hypothesizes that strategic online reputation-management has a positive influence on the strategic use of revenue-management. The results showed a statistically significant relationship between the two variables ($p < 0.01$) and a direct effect of 0.34, while the confidence interval was between 0.14 and 0.60. Thus, this hypothesis finds empirical support, and we can conclude that strategic online reputation-management exerts a positive impact on the strategic use of revenue-management.

In short, the results show that strategic online reputation-management has a positive effect on

Table 4. Discriminant validity.

Construct	Strategic online reputation-management	Strategic use of revenue-management	FPV	Online reputation
Strategic online reputation-management	I			
Strategic use of revenue-management	0.33 (0.13; 0.59) **	I		
FPV	0.48 (0.24; 0.68) **	0.51 (0.31; 0.69) **	I	
Online reputation	0.29 (0.09; 0.47) *	0.12 (−0.01; 0.24) n.s.	0.19 (0.05; 0.33) *	I

Standardized coefficients (confidence interval); ** $p \leq 0.01$; * $p \leq 0.05$.
 FPV=firm perceived value; n.s.=nonsignificant.

**Figure 3.** Results of hypotheses. Standardized coefficients (confidence interval); ** $p \leq 0.01$; * $p \leq 0.05$. FPV=firm perceived value.

the strategic use of revenue-management and that both strategies have a positive and significant effect on FPV, which, in turn, has a positive and significant effect on the firm's online reputation.

Discussion of the results, conclusions and business implications

For consumers, the content generated by other users in online media is a source of independent,

reliable information that they feel they can trust (Nieto-Garcia et al., 2019). Therefore, online reputation exerts a major influence on the consumer's purchase decision-making process (Bigné et al., 2020). Combining the internal perspective of the hotel firm with the external perspective of the market, the present research seeks to determine what strategic actions the firm might take to improve its online reputation.

To fulfill the research aim, we proposed and validated a model that captures the effect of FPV on online reputation. The study also

examines strategic online reputation-management and revenue-management—both of which leverage the market intelligence generated via online media—as antecedents of FPV, together with the relationship between the two strategies. The work makes several contributions to the literature, which we now describe.

First, FPV was found to exert a significant and positive effect on the firm's online reputation. Although the literature has shown that delivering superior value to the market has a positive effect on consumer behavior (e.g. Gallarza et al., 2015; Martelo-Landroque et al., 2015), this study responds to the future line of research proposed by Pérez-Aranda et al. (2018), addressing for the first time the relationship between the value delivered (from the firm's perspective) and the firm's online reputation (according to the market perspective). This finding is key, since it is the consideration of both perspectives that will achieve a positive effect on consumer behavior (Martelo-Landroque et al., 2015).

Second, the study demonstrates that strategic online reputation-management and revenue-management both exert a positive effect on FPV. Although the literature highlights, on the theoretical level, that having mechanisms in place to identify consumer needs and preferences (such as strategic online reputation-management) enables firms to offer superior value to the market (Nasution and Mavondo, 2008) and that the use of market intelligence (obtained via online media) enables revenue-management to offer superior value to consumers (Selmi and Chaney, 2018), the present work goes a step further by empirically demonstrating both relationships. Here, once more, it responds to the call for future lines of research proposed by authors including Bore et al. (2017), Denizci Guillet and Mohammed (2015), Niu and Fan (2018), and Selmi and Chaney (2018).

Third, our findings demonstrate that strategic online reputation-management has a positive and significant effect on the firm's strategic use of revenue-management. Although the literature has theoretically defended the proposition that strategic revenue-management relies on strategic online reputation-management (Alrawadieh et al., 2021), the present study shows this relationship empirically.

Finally, this study has shown that the value offered by the hotel firm to the market—drawing on the possibilities offered by market intelligence thanks to strategic online reputation-management and revenue-management—has a

positive effect on the market's evaluation of the firm, as expressed in its online reputation. These results constitute advancements in the literature on online reputation: (1) by investigating its antecedents, which scholars highlight as a necessary step (De Pelsmacker et al., 2018; Xu et al., 2019a) and (2) by ascertaining the firm's perspective (via FPV) and the market's perspective (via online reputation) and testing the effect of the former on the latter. This effect has not been addressed, to date, despite the fact that the literature contends that the firm's point of view should be taken into account in the study of online reputation (Baka, 2016; Bore et al., 2017; De Pelsmacker et al., 2018; Garrido-Moreno et al., 2018; Kwok et al., 2017).

Practical implications

The results of this research are of practical use to the professional sector as they help firms—in particular, hotel firms—to identify the strategic mechanisms that will enable them to secure a competitive advantage based on the enhancement of online reputation. Managers must be mindful that, to improve their firm's online reputation, they must focus on delivering superior value to consumers by ensuring that this value is based on what the consumer expects from the services it provides. For example, the hotel firm can analyze whether it is delivering an offer that the market considers good value-for-money, whether its offers are considered a good choice by consumers, and whether its prices are considered reasonable (Nasution and Mavondo, 2008).

The present study also demonstrates that it is important for firms to generate and exploit market intelligence obtained from online media, as this will enable them to deliver superior value to the market, implementing specific strategies, such as strategic online reputation-management and revenue-management. The results show that, to increase the value it delivers to the market, the firm must strategically manage its online reputation—that is, it must encourage consumers to comment on their experience of the firm in online media, analyze the customer ratings it receives on specialized websites, analyze whether the experiences shared by consumers are desirable to potential consumers, and communicate personalized responses to the comments posted by consumers on online media (De Pelsmacker et al., 2018; Marchioria et al., 2013). This study therefore demonstrates that hotel firms must harness online reputation-

management *strategically*, to ensure that its function is not limited to responding to consumer reviews but also seeks to actively encourage such reviews and leverage them as intelligence with which to make service improvements and deliver superior value to consumers. To obtain the optimal benefit from this strategy, hotel firms are recommended to: (1) use all possible avenues to encourage customers to write reviews (e.g. via Reception personnel, displaying a special QR code at the hotel with a link to independent review sites, or via email and social networks); (2) draw on the wide range of software options available, both free and paid, that provide real-time updates on consumer opinions (e.g. Google Alerts, Social Mention, or Trust You); and (3) implement procedures to ensure that these opinions are systematically classified, analyzed, and acted-upon, accordingly, either by simply responding to the comment or by also instigating service improvements.

Turning to revenue-management, its *strategic* use can contribute to delivering superior value to the market by optimizing market segmentation, ensuring dynamic capacity-management, and creating and managing demand for each market segment (Selmi and Chaney, 2018). Therefore, hotel firms are advised to position revenue-management as a strategic and not a tactical function if they wish to deliver greater value to the consumer. Furthermore, it is recommended that they take steps to convey to employees the importance of this function, so that the entire enterprise can steer the business toward achieving improved *long-term* performance, favoring this approach over short-term revenue. On this basis, an important practical implication of the present study derives from the finding that the strategic application of revenue-management must be supported by strategic online reputation-management. Therefore, those personnel tasked with managing the online marketing strategies of hotel firms must take advantage of the market intelligence that can be generated thanks to strategic online reputation-management, by integrating it into their revenue-management system. This constitutes a key step toward giving it a strategic role based on long-term planning of pricing and capacity, based on market intelligence.

In short, if firms are seeking to improve the online reputation they are achieving in the market, they should adopt approaches such as strategic online reputation-management and strategic revenue-management, as these will contribute to improving the value delivered to the

market. In turn, this will positively influence firms' online reputation.

Limitations and future research directions

Like all research, this study has limitations that may point to potential areas of research for the future. This study was based on a population comprising hotels belonging to hotel chains with a presence in Spain—a type of firm that is characterized by innovation in the development of online strategies. Hence, any generalization of the results to other profiles of firms should be interpreted with caution. A future line of research would be to replicate the study in other types of tourism firms and in other geographical areas. A further possible line of future research would be to include possible antecedents of strategic online reputation-management in the research model, such as customer relationship management or market or customer orientation.

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Supplemental material

Supplemental material for this article is available online.

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